ANSWER KEY

Quiz 06 - Polymorphism (Multiple Choice, Short)

Multiple choice. Choose the **one** best answer for each question and circle it on the **answer sheet** (page 5). All code is from Worksheet 06. Every snippet compiles unless a question says otherwise. About 10 minutes.

part 1 Static vs. Dynamic Binding

```
class Vehicle {
public:
    void describe() { cout << "A vehicle.\n"; }
};
class Car : public Vehicle {
public:
    void describe() { cout << "A car.\n"; }
};

int main() {
    Car c;
    Vehicle* vp = &c;      // base-class POINTER to a Car
    Vehicle& vr = c;       // base-class REFERENCE to a Car

    c.describe();          // (a)
    vp->describe();        // (b)
    vr.describe();         // (c)
}
```

1. As written, what does line (b) `vp->describe();` print?

   (A) `A car.`
   (B) `A vehicle.`
   (C) `A vehicle.` then `A car.`
   (D) Nothing. It's a compile error.

2. `Vehicle::describe()` is changed to `virtual void describe()`. Now what does line (c) `vr.describe();` print?

   (A) `A car.`
   (B) `A vehicle.`
   (C) `A vehicle.` then `A car.`
   (D) Nothing. It's a compile error.

3. Adding `virtual` changes exactly two of the three lines. Which call does *not* change?

   (A) (b) only
   (B) (c) only
   (C) None. All three change.
   (D) (a) only

4. Without `virtual`, the compiler picks which `describe()` to run from…

   (A) the actual object's type, at run time
   (B) the order the classes were declared
   (C) the pointer's or reference's declared type, at compile time
   (D) the most recently constructed object

5. Which call has the version that runs chosen at **run time**?

   (A) `for (auto& s : shapes) s->draw();` where `draw()` is `virtual`
   (B) `print(5);` where `print(int)` and `print(string)` both exist
   (C) `Fraction c = a + b;` using the `operator+` you wrote in W04
   (D) `Vehicle* v = new Car(); v->describe();` where `describe()` is **not** virtual

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part 2 `virtual`, `override`, and the Signature Rules

Treat each numbered line as the *only* function in `Wizard`.

```
class Character {
public:
    virtual ~Character() {}
    virtual void attack();
    virtual void printName() const;
    virtual void heal(int amount);
    void rest();                          // NOT virtual
};

class Wizard : public Character {
public:
    void attack() override;               // (1)
    void printName() override;            // (2)
    void heal(double amount) override;    // (3)
    void rest() override;                 // (4)
    void Attack() override;               // (5)
    void heal(int amount);                // (6)  no 'override' keyword
};
```

6. Why is line (2) a compile error?

- (A) `printName()` isn't virtual in `Character`
- (B) The name doesn't match
- (C) The parameter types don't match
- **(D) It's missing `const`**

7. Why is line (4) a compile error?

- (A) It compiles and overrides fine
- **(B) `rest()` isn't virtual in `Character`**
- (C) It's missing `const`
- (D) The parameter types don't match

8. You delete the word `override` from line (2), so it now compiles. Given `Character* p = new Wizard();`, which version does `p->printName();` run?

- **(A) `Character::printName()`. Wizard's version hides it instead of overriding it.**
- (B) `Wizard::printName()`
- (C) Both, base version first
- (D) It crashes at run time

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part 3 Pointers, References, and Slicing

```
class Character {
protected:
    string name;
public:
    Character(string n) : name(n) {}
    virtual ~Character() {}
    virtual void printName() { cout << name << endl; }
};
class Wizard : public Character {
public:
    Wizard(string n) : Character(n) {}
    void printName() override { cout << "The magnificent " << name << endl; }
};
class Warrior : public Character {
public:
    Warrior(string n) : Character(n) {}
    void printName() override { cout << "The powerful " << name << endl; }
};

void byRef(Character& c) { c.printName(); }
void byPtr(Character* c) { c->printName(); }
void byVal(Character c)  { c.printName(); }

int main() {
    Wizard  merlin("Merlin");
    Warrior conan("Conan");

    byRef(merlin);                                  // (a)
    byPtr(&conan);                                  // (b)
    byVal(merlin);                                  // (c)

    vector<Character>  copies = {merlin, conan};
    copies[1].printName();                          // (d)

    vector<Character*> party  = {&merlin, &conan};
    party[1]->printName();                          // (e)
}
```

9. What does line (c) print?

   (A) `The magnificent Merlin`
   (B) `The powerful Merlin`
   (C) Nothing. It's a compile error.
   (D) `Merlin`

10. What does line (d) print?

   (A) `The powerful Conan`
   (B) `Conan`
   (C) `Merlin`
   (D) `The magnificent Merlin`

11. What does line (e) print?

   (A) `Conan`
   (B) `The magnificent Merlin`
   (C) `The powerful Conan`
   (D) Nothing. It's a compile error.

12. Lines (c) and (d) both print the plain name even though `printName()` is virtual. What is the one-character change to `byVal` that fixes (c)?

   (A) Change the parameter to `const Character c`
   (B) Change the parameter to `Character& c`
   (C) Add `override` to `Character::printName()`
   (D) Change `c.printName()` to `c->printName()`

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part 4 Virtual Destructors

```
class Base {
public:
    ~Base() { cout << "~Base "; }                  // not virtual
};

class Derived : public Base {
    int* data;
public:
    Derived() : data(new int[100]) {}
    ~Derived() { cout << "~Derived "; delete[] data; }
};

int main() {
    Base* p = new Derived();
    delete p;
}
```

13. As written, what does the program print?

- (A) `~Derived ~Base`
- **(B) `~Base`**
- (C) `~Base ~Derived`
- (D) Nothing

14. The fix is one word. Which word, and where does it go?

- (A) `virtual` in front of `~Derived()`
- (B) `delete[]` in place of `delete` in `main()`
- **(C) `virtual` in front of `~Base()`**
- (D) `override` after `~Derived()`

15. After the fix, what does the program print?

- (A) `~Base ~Derived`
- (B) `~Base`
- (C) `~Derived`
- **(D) `~Derived ~Base`**

16. Finish the rule of thumb: “If a class has ____, give it a ____.”

- **(A) any virtual functions; virtual destructor**
- (B) a constructor; destructor
- (C) pointer members; const method
- (D) a base class; copy constructor

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answers **Answer Sheet**

circle one letter per question

1. A **B** C D
2. **A** B C D
3. A B C **D**
4. A B **C** D
5. **A** B C D
6. A B C **D**
7. A **B** C D
8. **A** B C D
9. A B C **D**
10. A **B** C D
11. A B **C** D
12. A **B** C D
13. A **B** C D
14. A B **C** D
15. A B C **D**
16. **A** B C D